



DOCUMENTATION

IndigoCorp Enterprise Network



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NETWORK OBJECTIVES

The primary objective of this project was to design and implement a secure, scalable enterprise network using Cisco Packet Tracer. The network was developed to simulate a realistic business environment connecting multiple departments, branch offices, servers, and wireless infrastructure.

The project aimed to achieve the following objectives:

- Design a small to medium-sized enterprise network architecture.
- Implement VLAN segmentation to improve security and reduce broadcast traffic.
- Configure Inter-VLAN Routing using a Layer 3 switch.
- Deploy OSPF as the dynamic routing protocol.
- Implement dual-stack IPv4 and IPv6 networking.
- Configure enterprise network services including DHCP, DNS and TFTP.
- Provide secure device management using SSH.
- Deploy enterprise wireless networking using a Wireless LAN Controller (WLC) and a Lightweight Access Point (LAP).
- Enhance my practical networking skills aligned with Cisco technologies

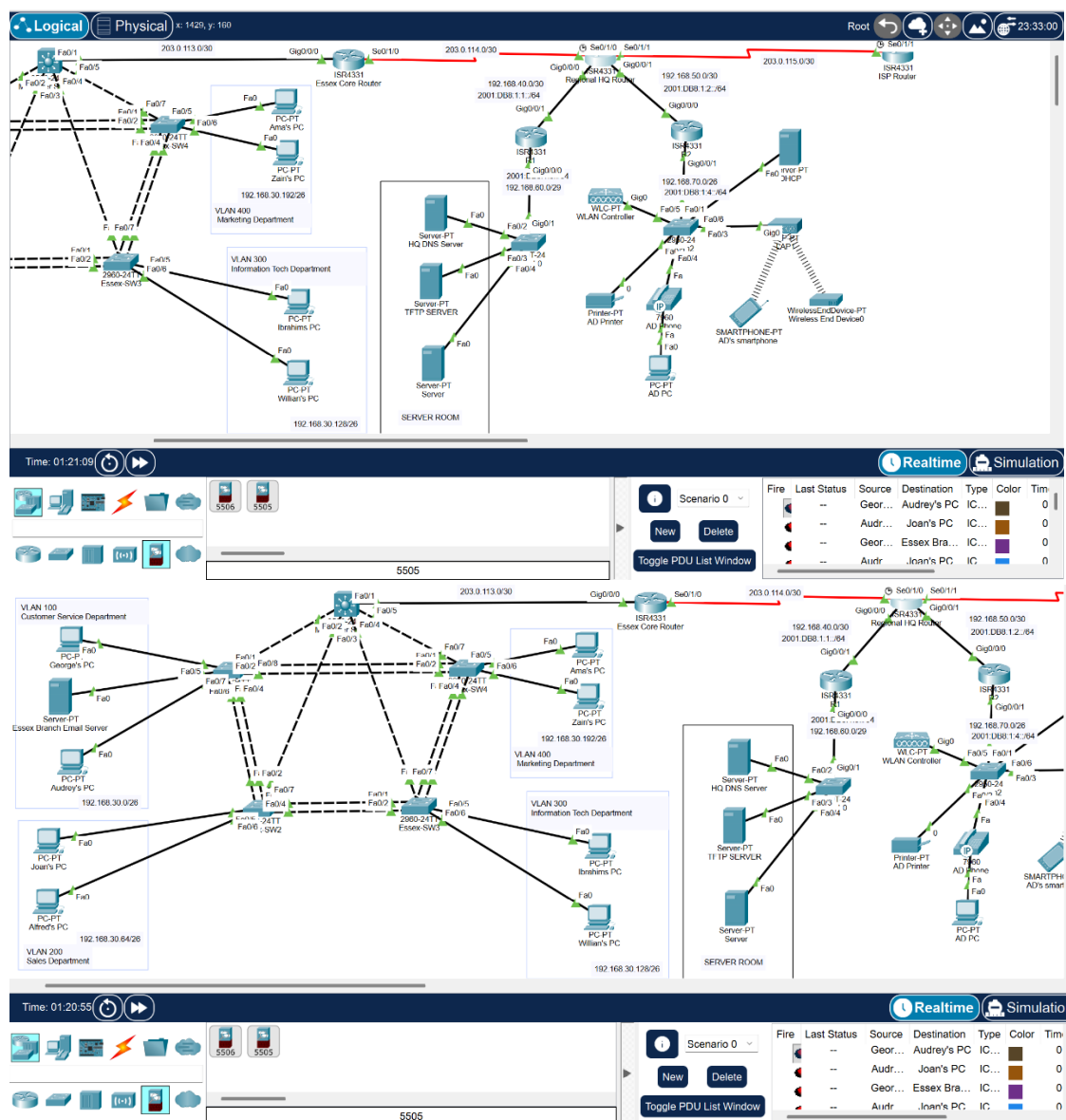
NETWORK TOPOLOGY

The enterprise network consists of two geographically separated sites connected through routed WAN links and a simulated Internet Service Provider (ISP) router.

The Essex Branch contains a Layer 3 core switch connected to multiple access switches that support the Customer Service, Sales, Information Technology, and Marketing departments.

The Regional Headquarters hosts the organisation's primary network infrastructure, including enterprise routers, DHCP, DNS and TFTP servers, wireless infrastructure, printers, IP telephony and end-user devices.

A dedicated ISP router provides simulated Internet connectivity to demonstrate WAN routing and NAT/PAT implementation.



NETWORK ARCHITECTURE

Essex Branch

The Essex Branch is built around a Layer 3 switch that performs inter-VLAN routing for departmental networks. Multiple Layer 2 access switches connect end-user devices to the enterprise network through IEEE 802.1Q trunk links. They are also interconnected and enabled with the link aggregation control protocol (EtherChannel) to provide redundancy.

Regional Headquarters

The Regional Headquarters contains the organisation's routing infrastructure, enterprise servers and wireless services. Two routers connect internal networks while DHCP, DNS and TFTP servers provide essential network services to the subnetwork.

VLAN DESIGN

VLAN	DEPARTMENT
10	Voice
80	Wireless
99	Management
100	Customer Service
200	Sales
300	Information Technology
400	Marketing

IP ADDRESSING

Network	Purpose
192.168.30.0/30	Allocated to the Customer Service Department
192.168.30.64/30	Allocated to the Sales Department
192.168.30.128/30	Allocated to the IT Department
192.168.30.192/30	Allocated to the Marketing Department
192.168.40.0/30	Regional HQ to Router R1
192.168.50.0/30	Regional HQ to Router R2
192.168.60.0/30	Server Network
192.168.70.0/30	Head Office LAN
203.0.113.0/30	Essex Layer 3 Switch to Essex Core Router
203.0.114.0/30	Essex Core Router to Regional HQ
203.0.115.0/30	Regional HQ to ISP

ROUTING DESIGN

Open Shortest Path First (OSPF) was selected as the enterprise routing protocol due to its scalability, fast convergence and widespread use in modern enterprise environments.

All routers participate in OSPF Area 0 and advertise directly connected networks. Loopback interfaces were configured to provide stable router IDs.

Static default routes were configured to forward unknown traffic towards the Regional Headquarters and ISP router, allowing enterprise devices to access simulated Internet resources.

The Layer 3 switch performs Inter-VLAN routing, enabling communication between departmental VLANs while maintaining network segmentation.

WIRELESS INFRASTRUCTURE

Enterprise wireless networking was implemented using Cisco wireless technologies.

The wireless infrastructure consists of:

- Wireless LAN Controller (WLC)
- Lightweight Access Point (LAP)
- Wireless VLAN
- Wireless Smartphone Client and Wireless End Device

The Wireless LAN Controller centrally manages the Lightweight Access Point, enabling wireless users to securely connect to the enterprise network.

Wireless clients automatically receive network configuration from the DHCP server and can communicate with internal network resources through the wireless VLAN.

This implementation reflects the centralised wireless architecture commonly deployed in enterprise environments.

SECURITY FEATURES

Implemented security measures include:

- Secure Shell (SSH) Version 2 for encrypted remote management.
- RSA key generation for secure authentication.
- Password encryption to protect locally configured credentials.
- Port Security using Sticky MAC addresses to prevent unauthorised devices from connecting.
- Dedicated Management VLAN for administrative access.
- Shutdown of unused switch ports to reduce attack surfaces.
- VLAN segmentation to isolate departmental traffic.

These features provide a more secure enterprise environment while demonstrating industry-standard security practices.

TESTING AND VALIDATION

Successful validation included:

- OSPF neighbour relationships established successfully.
- Successful Inter-VLAN communication.
- Dynamic IP address allocation through DHCP.
- Successful DNS name resolution.
- Wireless client connectivity through the Wireless LAN Controller.
- NAT/PAT translation through the ISP router.
- SSH remote management of network devices.
- Successful IPv4 and IPv6 connectivity.

AD PC

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.70.16

Subnet Mask

255.255.255.192

Default Gateway

192.168.70.1

DNS Server

192.168.60.2

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::207:ECFF:FE0E:33A1

Default Gateway

DNS Server

802.1X

Use 802.1X Security

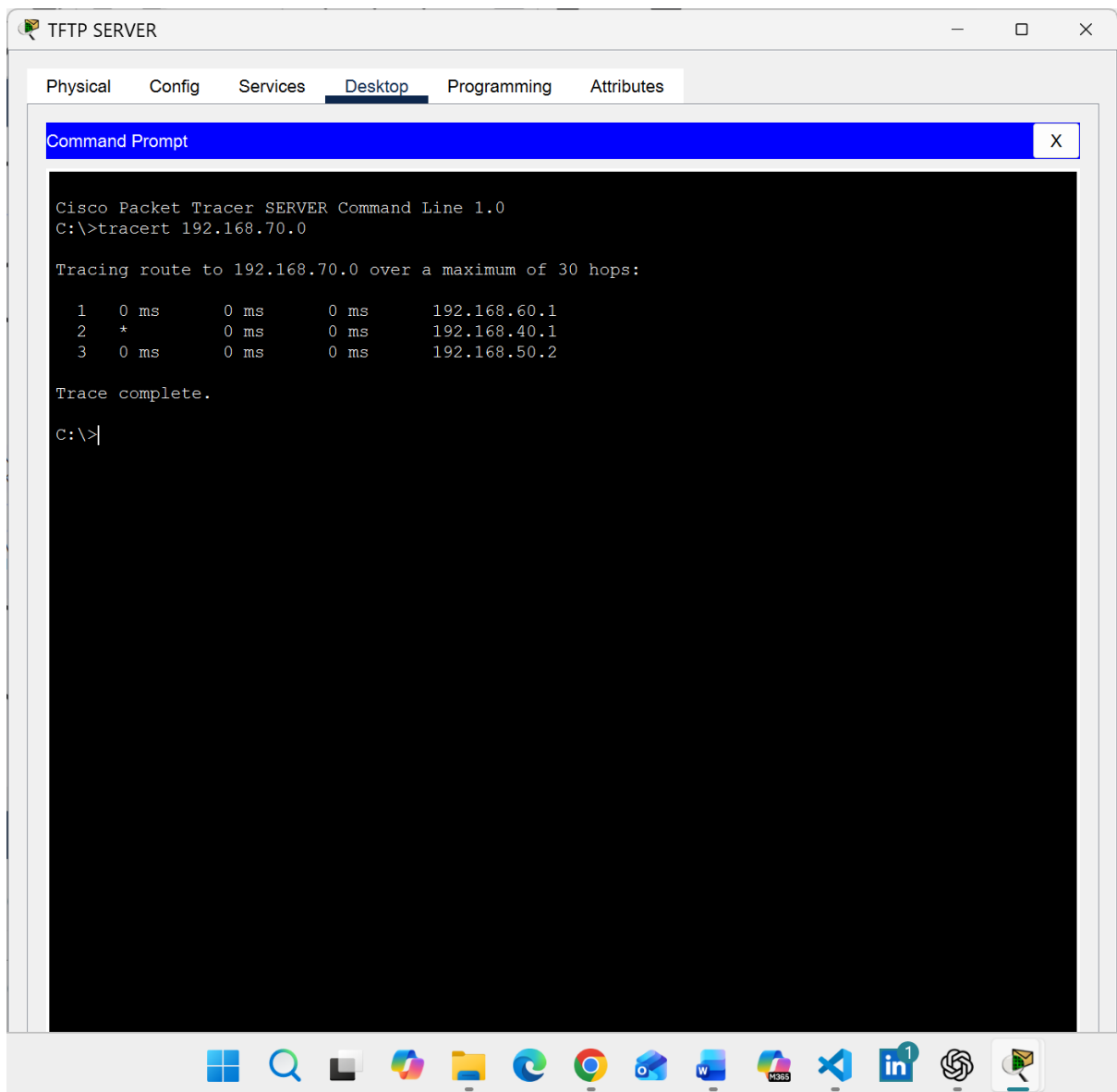
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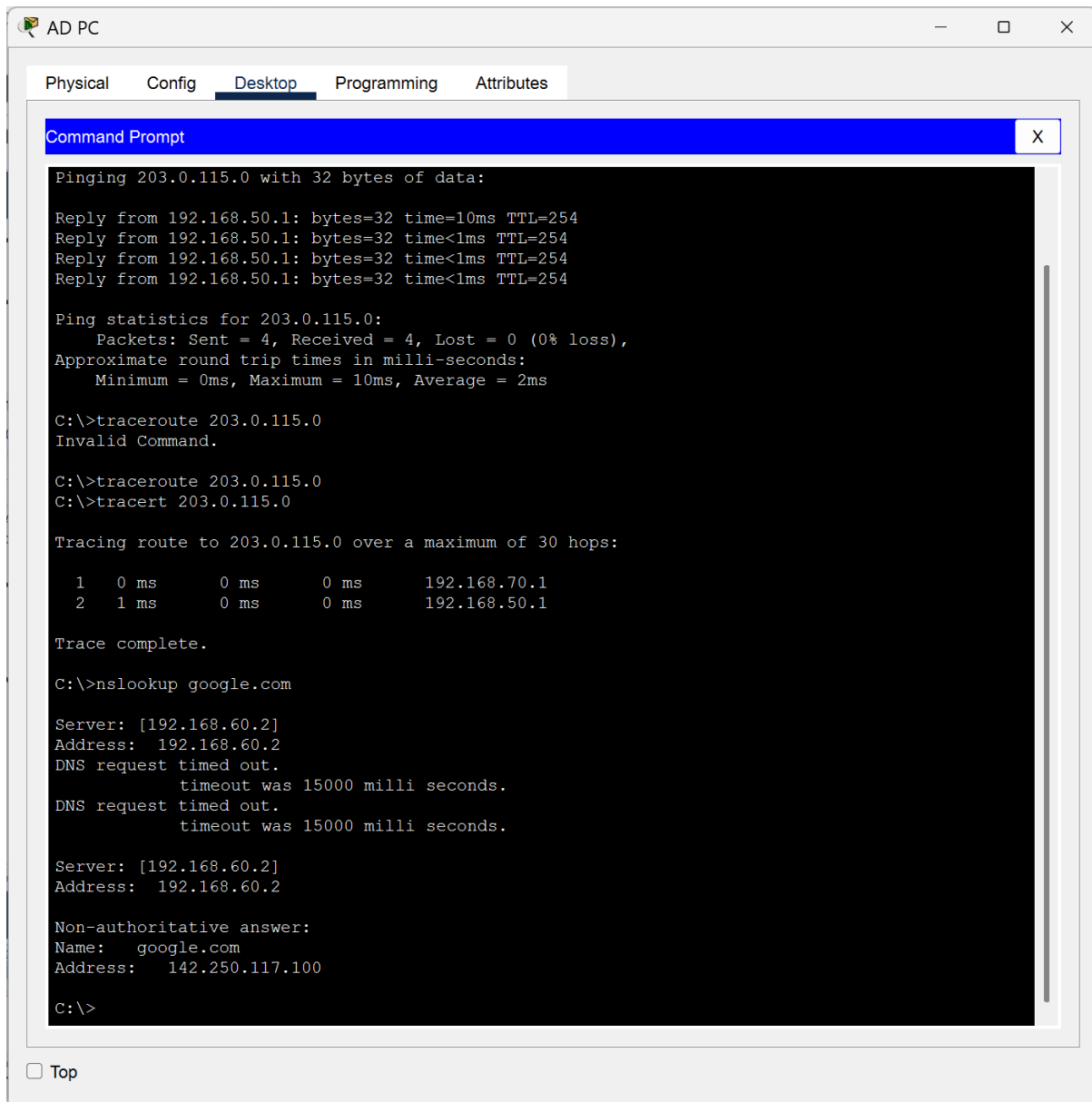
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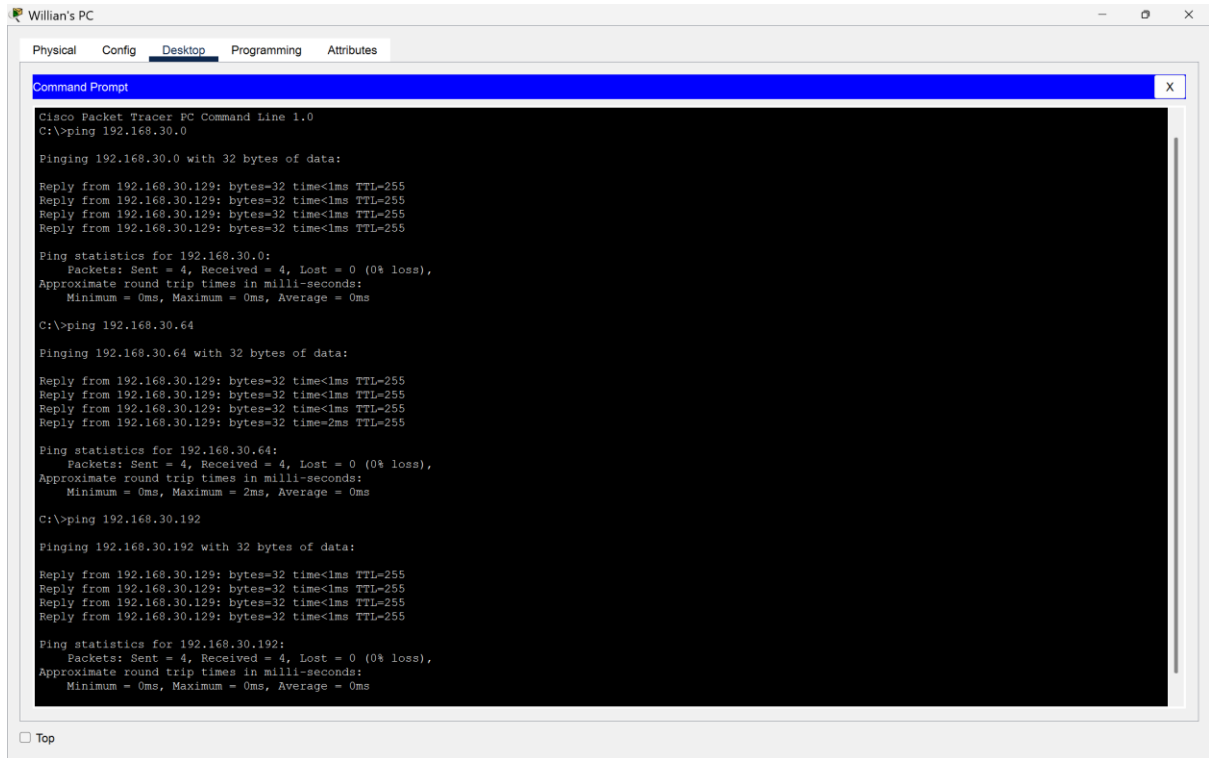
Username

Password

Top







CONCLUSION

This project successfully demonstrates the design and implementation of a modern enterprise network using Cisco Packet Tracer. The completed network integrates routing, switching, wireless networking, IPv4/IPv6 dual-stack addressing and enterprise services into a scalable and secure architecture.

Through this project, practical experience was gained in designing and configuring enterprise network infrastructure using Cisco technologies.